Run report: itemOJ-run-20261018T2124Z

Descriptive record of one recorded run of examples/n2_routes.rs, mode run (Parts O/J) under `research/weil_descent_angles_20261018/PROTOCOL.md` (Addendum A and Amendments B–G, each committed before the run it governs). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, 4 threads); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
commit 4396b057a1a918da5c98bbbed0b10d5d04776a75
cmd: n2 research/weil_descent_angles_20261018/results/itemOJ-run-20261018T2124Z check; n2 research/weil_descent_angles_20261018/results/itemOJ-run-20261018T2124Z run (N2_THREADS=4, sizes 20..40)
rustc 1.97.0 (2d8144b78 2026-07-07)
4
Sun Oct 18 21:24:50 UTC 2026
post-run (d2edb2ab8): the tool was rustfmt-formatted (format only; unit tests pass); outputs unaffected.
```

tool_log.txt

```
jobs 18 total 57.6s
```

checks.txt

```
K1 decomposition oracle vs brute force (p <= 101): instances 4, points 6068, mismatches 0
K2-K4 p=263 N=69544 h=8 ell=8693: #C(F_p)=264 (p+1=264) | ell·P=O true | tau valid true hom true nonzero true ell·tau(P)=0 true | N·D=0 true
K2-K4 p=271 N=73624 h=8 ell=9203: #C(F_p)=272 (p+1=272) | ell·P=O true | tau valid true hom true nonzero true ell·tau(P)=0 true | N·D=0 true
K2-K4 p=1097 N=1205332 h=4 ell=301333: #C(F_p)=1098 (p+1=1098) | ell·P=O true | tau valid true hom true nonzero true ell·tau(P)=0 true | N·D=0 true
K2-K4 p=1069 N=1142056 h=8 ell=142757: #C(F_p)=1070 (p+1=1070) | ell·P=O true | tau valid true hom true nonzero true ell·tau(P)=0 true | N·D=0 true
K2-K4 p=4229 N=17878244 h=4 ell=4469561: #C(F_p)=4230 (p+1=4230) | ell·P=O true | tau valid true hom true nonzero true ell·tau(P)=0 true | N·D=0 true
K2-K4 p=4507 N=20306164 h=4 ell=5076541: #C(F_p)=4508 (p+1=4508) | ell·P=O true | tau valid true hom true nonzero true ell·tau(P)=0 true | N·D=0 true
K5 p=1123: genus-2 add fast 41.3M+1.01I cantor 112.8M+4.00I | double fast 48.0M+1.00I cantor 115.9M+3.99I | agree true | E(F_p^2) add 12M+1I
K5 p=4561: genus-2 add fast 41.0M+1.00I cantor 113.0M+4.00I | double fast 48.0M+1.00I cantor 116.0M+4.00I | agree true | E(F_p^2) add 12M+1I
K5 p=18251: genus-2 add fast 41.0M+1.00I cantor 113.0M+4.00I | double fast 48.0M+1.00I cantor 116.0M+4.00I | agree true | E(F_p^2) add 12M+1I
K5 p=71963: genus-2 add fast 41.0M+1.00I cantor 113.0M+4.00I | double fast 48.0M+1.00I cantor 116.0M+4.00I | agree true | E(F_p^2) add 12M+1I
```

summary.txt

```
O/J bits 20 arm J c 1: IC/rho ratio in [0.507, 1.542] over 12 cells -> unresolved; steps/p 0.311 (rho 0.338)
O/J bits 20 arm J c 2: IC/rho ratio in [0.508, 1.533] over 12 cells -> unresolved; steps/p 0.311 (rho 0.338)
O/J bits 20 arm J c 4: IC/rho ratio in [0.501, 1.488] over 12 cells -> unresolved; steps/p 0.241 (rho 0.338)
O/J bits 20 arm D c 1: IC/rho ratio in [2.950, 4.183] over 6 cells -> rho cheaper in every cell; steps/p 0.397 (rho 0.338)
O/J bits 20 arm D c 2: IC/rho ratio in [2.489, 3.491] over 6 cells -> rho cheaper in every cell; steps/p 0.320 (rho 0.338)
O/J bits 20 arm D c 4: IC/rho ratio in [3.368, 4.806] over 6 cells -> rho cheaper in every cell; steps/p 0.457 (rho 0.338)
O/J bits 24 arm J c 1: IC/rho ratio in [0.661, 1.954] over 12 cells -> unresolved; steps/p 0.351 (rho 0.283)
O/J bits 24 arm J c 2: IC/rho ratio in [0.680, 1.995] over 12 cells -> unresolved; steps/p 0.364 (rho 0.283)
O/J bits 24 arm J c 4: IC/rho ratio in [0.659, 1.899] over 12 cells -> unresolved; steps/p 0.321 (rho 0.283)
O/J bits 24 arm D c 1: IC/rho ratio in [9.111, 15.079] over 6 cells -> rho cheaper in every cell; steps/p 0.461 (rho 0.283)
O/J bits 24 arm D c 2: IC/rho ratio in [9.781, 16.216] over 6 cells -> rho cheaper in every cell; steps/p 0.498 (rho 0.283)
O/J bits 24 arm D c 4: IC/rho ratio in [6.910, 11.350] over 6 cells -> rho cheaper in every cell; steps/p 0.339 (rho 0.283)
O/J bits 28 arm J c 1: IC/rho ratio in [0.707, 2.119] over 12 cells -> unresolved; steps/p 0.338 (rho 0.425)
O/J bits 28 arm J c 2: IC/rho ratio in [0.681, 2.036] over 12 cells -> unresolved; steps/p 0.320 (rho 0.425)
O/J bits 28 arm J c 4: IC/rho ratio in [0.700, 2.095] over 12 cells -> unresolved; steps/p 0.330 (rho 0.425)
O/J bits 28 arm D c 1: IC/rho ratio in [12.384, 23.683] over 6 cells -> rho cheaper in every cell; steps/p 0.357 (rho 0.425)
O/J bits 28 arm D c 2: IC/rho ratio in [13.013, 24.874] over 6 cells -> rho cheaper in every cell; steps/p 0.376 (rho 0.425)
O/J bits 28 arm D c 4: IC/rho ratio in [13.429, 25.760] over 6 cells -> rho cheaper in every cell; steps/p 0.379 (rho 0.425)
O/J bits 32 arm J c 1: IC/rho ratio in [0.900, 2.774] over 12 cells -> unresolved; steps/p 0.348 (rho 0.378)
O/J bits 32 arm J c 2: IC/rho ratio in [0.701, 2.143] over 12 cells -> unresolved; steps/p 0.262 (rho 0.378)
O/J bits 32 arm J c 4: IC/rho ratio in [0.888, 2.728] over 12 cells -> unresolved; steps/p 0.339 (rho 0.378)
O/J bits 32 arm D c 1: IC/rho ratio in [17.961, 38.797] over 6 cells -> rho cheaper in every cell; steps/p 0.331 (rho 0.378)
O/J bits 32 arm D c 2: IC/rho ratio in [20.006, 43.090] over 6 cells -> rho cheaper in every cell; steps/p 0.371 (rho 0.378)
O/J bits 32 arm D c 4: IC/rho ratio in [20.953, 45.133] over 6 cells -> rho cheaper in every cell; steps/p 0.389 (rho 0.378)
O/J bits 36 arm J c 1: IC/rho ratio in [0.589, 1.852] over 12 cells -> unresolved; steps/p 0.343 (rho 0.626)
O/J bits 36 arm J c 2: IC/rho ratio in [0.596, 1.874] over 12 cells -> unresolved; steps/p 0.347 (rho 0.626)
O/J bits 36 arm J c 4: IC/rho ratio in [0.566, 1.778] over 12 cells -> unresolved; steps/p 0.328 (rho 0.626)
O/J bits 36 arm D c 1: IC/rho ratio in [16.647, 40.044] over 6 cells -> rho cheaper in every cell; steps/p 0.418 (rho 0.626)
O/J bits 36 arm D c 2: IC/rho ratio in [16.535, 39.834] over 6 cells -> rho cheaper in every cell; steps/p 0.411 (rho 0.626)
O/J bits 36 arm D c 4: IC/rho ratio in [15.745, 37.966] over 6 cells -> rho cheaper in every cell; steps/p 0.393 (rho 0.626)
O/J bits 40 arm J c 1: IC/rho ratio in [0.528, 1.680] over 12 cells -> unresolved; steps/p 0.394 (rho 0.812)
```

```
O/J bits 40 arm J c 2: IC/rho ratio in [0.410, 1.301] over 12 cells -> unresolved; steps/p 0.305 (rho 0.812)
O/J bits 40 arm J c 4: IC/rho ratio in [0.370, 1.169] over 12 cells -> unresolved; steps/p 0.274 (rho 0.812)
O/J bits 40 arm D c 1: IC/rho ratio in [12.866, 33.867] over 6 cells -> rho cheaper in every cell; steps/p 0.379 (rho 0.812)
O/J bits 40 arm D c 2: IC/rho ratio in [12.720, 33.332] over 6 cells -> rho cheaper in every cell; steps/p 0.381 (rho 0.812)
O/J bits 40 arm D c 4: IC/rho ratio in [13.185, 34.676] over 6 cells -> rho cheaper in every cell; steps/p 0.391 (rho 0.812)
O/J scaling arm R c 0: cost ~ p^0.787
O/J scaling arm J c 1: cost ~ p^0.792
O/J scaling arm J c 2: cost ~ p^0.762
O/J scaling arm J c 4: cost ~ p^0.758
O/J scaling arm D c 1: cost ~ p^1.072
O/J scaling arm D c 2: cost ~ p^1.086
O/J scaling arm D c 4: cost ~ p^1.076
O/J all answers verified: true
```

ranking.csv (first 12 of 36 rows)

bits	arm	c	cells	ic_cheaper_cells	rho_cheaper_cells	min_ratio	max_ratio	verdict
20	J	1	12	6	6	0.5069	1.5417	unresolved
20	J	2	12	6	6	0.5077	1.5327	unresolved
20	J	4	12	6	6	0.5007	1.4878	unresolved
20	D	1	6	0	6	2.9501	4.1828	rho cheaper in every cell
20	D	2	6	0	6	2.4895	3.4906	rho cheaper in every cell
20	D	4	6	0	6	3.3680	4.8059	rho cheaper in every cell
24	J	1	12	6	6	0.6609	1.9539	unresolved
24	J	2	12	6	6	0.6803	1.9946	unresolved
24	J	4	12	6	6	0.6595	1.8993	unresolved
24	D	1	6	0	6	9.1107	15.0789	rho cheaper in every cell
24	D	2	6	0	6	9.7812	16.2163	rho cheaper in every cell
24	D	4	6	0	6	6.9102	11.3495	rho cheaper in every cell

routes.csv (first 12 of 378 rows)

bits	instance	p	N	h	ell	target	arm	c	factor_base	steps	relations	b_relations	jac_ops	walk_M	walk_I	walk_native	jac_M	jac_I	la_M	la_I	la_native	eadd_M	eadd_I	verified	seconds
20	0	1091	1190984	8	148873	0	R	0	0	291	0	0	0	49105	3270	312	0	0	0	0	0	12	1	1	0.001
20	0	1091	1190984	8	148873	0	J	1	34	281	151	1	1941	89746	2048	942	87753	2025	0	0	1045	12	1	1	0.001
20	0	1091	1190984	8	148873	0	D	1	34	607	212	2	0	322779	7494	2928	0	0	0	0	1594	12	1	1	0.003
20	0	1091	1190984	8	148873	0	J	2	67	184	114	4	1755	81811	1860	724	79867	1837	0	0	688	12	1	1	0.001
20	0	1091	1190984	8	148873	0	D	2	67	381	166	6	0	214174	5420	2289	0	0	0	0	1135	12	1	1	0.002
20	0	1091	1190984	8	148873	0	J	4	133	200	104	6	1812	84998	1910	875	82228	1887	0	0	560	12	1	1	0.001
20	0	1091	1190984	8	148873	0	D	4	133	369	149	13	0	212847	5377	2372	0	0	0	0	815	12	1	1	0.003
20	0	1091	1190984	8	148873	1	R	0	0	317	0	0	0	49931	3318	358	0	0	0	0	0	12	1	1	0.001
20	0	1091	1190984	8	148873	1	J	1	34	356	182	3	1948	87715	1934	1146	85497	1911	0	0	1264	12	1	1	0.001
20	0	1091	1190984	8	148873	1	D	1	34	259	92	1	0	151137	4086	1345	0	0	0	0	552	12	1	1	0.001
20	0	1091	1190984	8	148873	1	J	2	67	244	123	2	1804	81555	1787	875	79370	1764	0	0	752	12	1	1	0.001
20	0	1091	1190984	8	148873	1	D	2	67	251	99	3	0	150811	4114	1519	0	0	0	0	622	12	1	1	0.001

routes_summary.csv (first 12 of 324 rows)

bits	arm	c	ipr	jac_pricing	native_w	runs	verified	mean_p	mean_cost	mean_rho_cost	ratio_to_rho	mean_steps_over_p	rho_steps_over_p
20	J	1	3M	measured	0	9	1	1081	94779.2	62581.1	1.5145	0.3112	0.3379
20	J	1	3M	measured	0.1	9	1	1081	95011.3	62621.4	1.5172	0.3112	0.3379
20	J	1	3M	measured	1	9	1	1081	97099.9	62984.0	1.5417	0.3112	0.3379
20	J	1	3M	eadd	0	9	1	1081	31725.0	62581.1	0.5069	0.3112	0.3379
20	J	1	3M	eadd	0.1	9	1	1081	31957.1	62621.4	0.5103	0.3112	0.3379
20	J	1	3M	eadd	1	9	1	1081	34045.7	62984.0	0.5405	0.3112	0.3379
20	J	1	2log2p_M	measured	0	9	1	1081	128568.6	122380.8	1.0506	0.3112	0.3379
20	J	1	2log2p_M	measured	0.1	9	1	1081	128800.7	122421.1	1.0521	0.3112	0.3379
20	J	1	2log2p_M	measured	1	9	1	1081	130889.3	122783.7	1.0660	0.3112	0.3379
20	J	1	2log2p_M	eadd	0	9	1	1081	65908.6	122380.8	0.5386	0.3112	0.3379
20	J	1	2log2p_M	eadd	0.1	9	1	1081	66140.7	122421.1	0.5403	0.3112	0.3379
20	J	1	2log2p_M	eadd	1	9	1	1081	68229.3	122783.7	0.5557	0.3112	0.3379

Graph

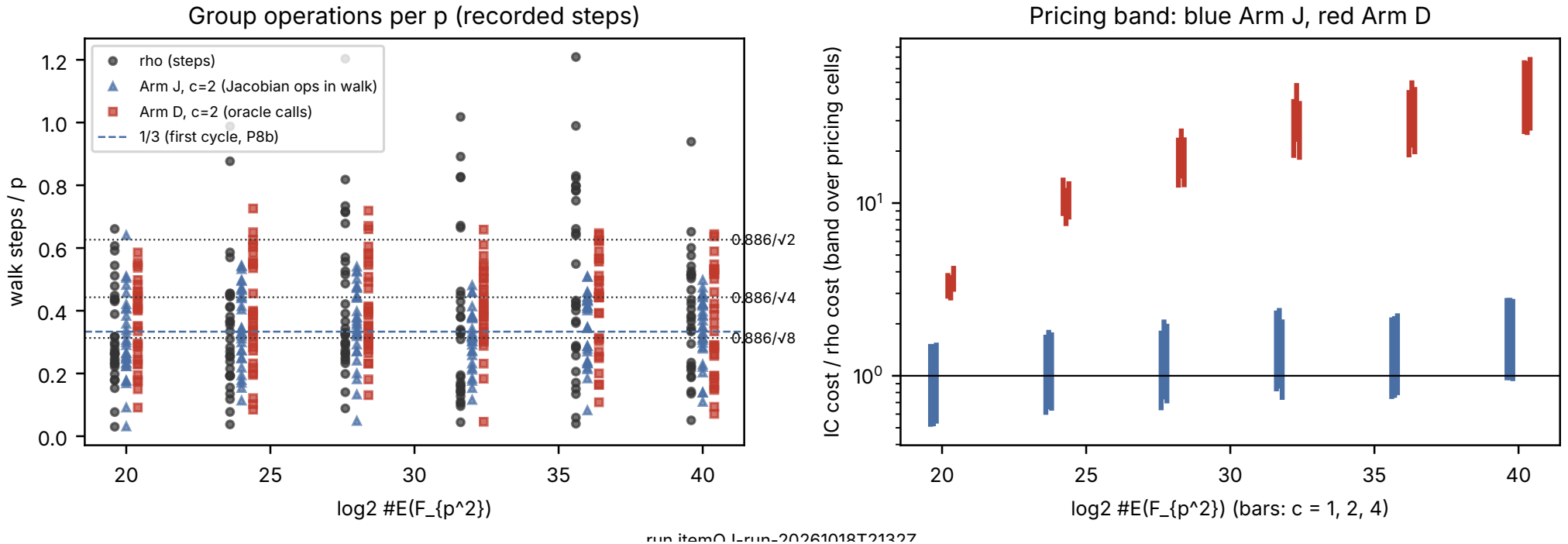


Figure 1: Graph drawn from the recorded re-run 20261018T2132Z (this run's rho rows are superseded by Amendment G); steps/p and pricing bands.